WORKED SOLUTION

STEP III 2005, Question 10

Mechanics • Elastic Band and Friction

STEP III 2005, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q10

The key idea

Everything depends on one variable: the distance d between the centres. The band's length is fixed by d, so its tension and stored energy are simple functions of d. From there, energy (elastic energy against work done by friction) tells us where the discs stop, and comparing the band's pull with the limiting friction tells us whether they stay stopped.

Solution

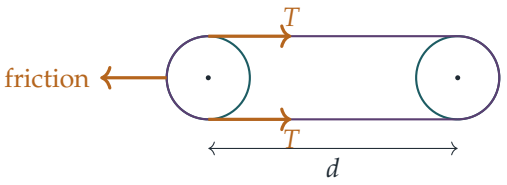
We model the discs as identical, the band as light and elastic with smooth contact, and the friction as having magnitude μmg whenever a disc slides. By symmetry the two discs always move equal distances in opposite directions, so the straight sections of the band stay parallel.

The band as a function of d. When the centres are a distance d apart, the band runs half-way round each disc and along two straight sections of length d. So its length is $2\pi r + 2d$ and its extension is $x = 2d$. By Hooke's law, with modulus $\lambda = \frac{\pi mg}{12}$ and natural length $2\pi r$,

$$T = \frac{\lambda x}{2\pi r} = \frac{\pi mg}{12} \cdot \frac{2d}{2\pi r} = \frac{mg\,d}{12r}, \qquad E = \frac{\lambda x^2}{2 \cdot 2\pi r} = \frac{mg\,d^2}{12r}.$$

Because the contact is smooth, the tension is the same all along the band. Each disc is pulled by the two straight sections, both parallel to the line of centres and pointing towards the other disc. So the resultant pull on each disc is

$$P = 2T = \frac{mg\,d}{6r}.$$



The figure, seen from above, shows the horizontal forces on the left-hand disc while it slides towards the other disc. Its weight and the normal reaction are vertical and cancel.

At release $d = 6r$, so $E_0 = 3mgr$ and $P = mg$. At contact $d = 2r$, so $E = \frac{1}{3}mgr$.

Energy while the discs approach. While the discs move inwards from $d = 6r$, each moves a distance $\frac{1}{2}(6r - d)$ against friction μmg. So the total work done against friction is $\mu mg(6r - d)$. The total kinetic energy is therefore

$$\begin{aligned} K(d) &= 3mgr - \frac{mg\,d^2}{12r} - \mu mg(6r - d) \\ &= mg\,(6r - d)\left(\frac{6r + d}{12r} - \mu\right), \end{aligned}$$

using $3r - \frac{d^2}{12r} = \frac{(6r-d)(6r+d)}{12r}$.

Part (i)

Starting. At release the pull on each disc is $P = mg$ and friction can supply at most μmg. So the discs start sliding exactly when $mg > \mu mg$, that is, when $\mu < 1$.

Stopping. Suppose $\mu < 1$. For $d < 6r$, $K(d)$ vanishes only where $\frac{6r+d}{12r} = \mu$, that is, at

$$d^* = 12\mu r - 6r.$$

As d decreases from $6r$, the second factor $\frac{6r+d}{12r} - \mu$ starts at $1 - \mu > 0$ and decreases, so $K > 0$ until d reaches d^*. The discs come to rest before colliding exactly when d^* is reached before contact, that is, when $2r < d^* < 6r$:

$$8r < 12\mu r < 12r \iff \tfrac{2}{3} < \mu < 1.$$

If $\mu \le \frac{2}{3}$, then $K(2r) \ge 0$ and the discs reach each other first. (When $\mu = \frac{2}{3}$ they arrive just touching with zero speed, which is not "before".)

> **Ask yourself** Once the discs stop at d^*, might the band start them moving again?

At d^* the pull is $P = \frac{mg\,d^*}{6r} = (2\mu - 1)mg$. This is less than the limiting friction μmg because $\mu < 1$, so static friction holds the discs and they stay at rest. Combining the two conditions, the discs start sliding but come to rest before colliding if and only if $\frac{2}{3} < \mu < 1$. ■

Part (ii)

If the discs collide at least once, they must reach $d = 2r$ for the first time moving inwards, with the approach formula still valid. So the total kinetic energy just before the first collision is

$$K(2r) = mg\,(4r)\left(\frac{8r}{12r} - \mu\right) = 4mgr\left(\tfrac{2}{3} - \mu\right) = \tfrac{4}{3}mgr(2 - 3\mu),$$

as required. (This is positive precisely when $\mu < \frac{2}{3}$, consistent with part (i).)

Part (iii)

First, $\mu^2 < \frac{4}{9}$ gives $\mu < \frac{2}{3}$, so by parts (i) and (ii) the discs do collide. Half the kinetic energy is lost, so they rebound from $d = 2r$ with total kinetic energy

$$K_1 = \tfrac{2}{3}mgr(2 - 3\mu).$$

Now the discs move apart. The band and friction both act towards the other disc, so each disc decelerates. While the separation increases from $2r$ to d, the elastic energy rises by $\frac{mg(d^2-4r^2)}{12r}$ and the work done against friction is $\mu mg(d - 2r)$, so

$$K(d) = K_1 - \frac{mg(d^2 - 4r^2)}{12r} - \mu mg(d - 2r).$$

This is strictly decreasing in d, so it vanishes exactly once, at the separation d_1 where the discs come to rest. Writing $d_1 = 2r + u$ and multiplying by $\frac{12r}{mg}$:

$$u^2 + 4r(1 + 3\mu)\,u - 8r^2(2 - 3\mu) = 0.$$

Taking the positive root and simplifying the discriminant, $(1 + 3\mu)^2 + 2(2 - 3\mu) = 9\mu^2 + 5$, gives

$$u = 2r\left(\sqrt{9\mu^2 + 5} - (1 + 3\mu)\right), \qquad d_1 = 2r\left(\sqrt{9\mu^2 + 5} - 3\mu\right).$$

(One can check that $2r < d_1 \le 6r$, so the band stays taut and the formula for its energy still applies.)

> **Ask yourself** The discs are momentarily at rest at d_1. What decides whether they stay there?

They stay at rest if the band's pull does not exceed the limiting friction:

$$\frac{mg\,d_1}{6r} \le \mu mg \iff \sqrt{9\mu^2 + 5} - 3\mu \le 3\mu \iff 9\mu^2 + 5 \le 36\mu^2 \iff \mu^2 \ge \tfrac{5}{27},$$

where squaring is valid because both sides are positive. So if $\frac{5}{27} < \mu^2 < \frac{4}{9}$, the discs collide, rebound, decelerate monotonically to rest at d_1, and are then held there by static friction for good. They come to rest exactly once after the first collision. ■

Checks

- At $\mu = \frac{2}{3}$ both part (i)'s stopping point $d^* = 12\mu r - 6r$ and part (ii)'s kinetic energy give "arrive at $d = 2r$ with zero speed", so the two parts agree at the boundary.
- With $\mu = 0.5$ (inside the range of part (iii)): $K(2r) = \frac{2}{3}mgr$, $K_1 = \frac{1}{3}mgr$ and $d_1 = 2r(\sqrt{7.25} - 1.5) \approx 2.385r$. The pull there is $\approx 0.397mg < 0.5mg$, so the discs stay put. A step-by-step numerical simulation of the motion reproduced all of this and also the stopping point $d^* = 12\mu r - 6r$ for several $\mu \in (\frac{2}{3}, 1)$. For $\mu^2 < \frac{5}{27}$ (for example $\mu = 0.42$) it showed the discs moving again after stopping, as predicted.

Common slips

- Taking the extension as d instead of $2d$: both straight sections stretch.
- Using only one tension on each disc: the band pulls each disc with *two* strands, giving $2T$.
- Forgetting to check that the discs stay at rest once stopped. "Speed zero" and "at rest for good" are different conditions.

Beyond STEP

Friction makes this a piecewise system. The equation of motion changes whenever the velocity changes sign, and "stuck" states appear where the applied force is below the friction threshold. Systems like this (stick–slip motion) are studied with phase-plane methods in first-year dynamics. For small μ, repeated collisions with energy loss produce infinitely many bounces in a finite time, which is worth thinking about. For more on energy methods and friction, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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